

# Kaleb Miller

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## Summary

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Senior Data Engineer with over 9 years of experience architecting and implementing enterprise data strategies, specializing in Azure and Spark-based solutions.

Demonstrated leadership in designing scalable, reliable, and secure data engineering platforms, with deep expertise in dimensional modeling, data lakes, and data warehouses.

Proven ability to lead and mentor teams, drive adoption of best practices, and deliver high-quality analytics, migrations, and data integrations.

Adept at collaborating with cross-functional stakeholders to deliver robust data solutions aligned with business objectives.

Experienced in supporting Microsoft ecosystem initiatives and fostering a culture of continuous learning.

## Education

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**Carnegie Mellon University**, Pittsburgh, PA

Master's degree, Data Science, August 2016 - May 2018

**Pennsylvania Western University**, California, PA

Bachelor's degree, Computer Science, September 2008 - June 2012

## Soft Skills

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- Effective Communication and Articulation
- Teamwork and Collaboration
- Empathy and Emotional Intelligence
- Creativity and Innovation
- Continuous Learning and Skill Development
- Client-Centric Approach and Customer Service
- Analytical mind

## Technical Skills

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- **Cloud Platforms:** Azure Data Lake Storage, AWS (S3, Glacier, EBS, EC2, Redshift, Elastic Beanstalk, CloudFormation), GCP BigQuery
- **Big Data & Processing:** Apache Spark, PySpark, Spark SQL, Hive, Pig, Databricks
- **Programming:** SQL, Python, PL/SQL, Shell, Node.js, PHP
- **Databases & Warehouses:** SQL Server, Snowflake, Redshift, BigQuery, PostgreSQL, MongoDB, Redis, Elasticsearch
- **Orchestration & Ingestion:** Apache Airflow, Apache Kafka, Apache NiFi
- **ETL / ELT & Transformation:** DBT, Talend, Incremental Models, Materializations, Python ETL
- **Data Modeling:** Dimensional modeling, Schema Design, Data Marts, Analytical Data Stores
- **Containers & DevOps:** Docker, Kubernetes, Ansible, Git-based CI/CD

- **APIs & Services:** Flask, Django, SQLAlchemy, REST APIs
- **Monitoring & Logging:** Elasticsearch, Log Indexing, Custom Monitoring Scripts
- **BI & Analytics:** Analytical Reporting, Python, Pandas
- **Infrastructure & Reliability:** Backup, Replication, Failover, Lifecycle Policies
- **Extra:** Azure, Data Factory, ADLS, Azure Databricks, Spark, Synapse Analytics, Azure SQL Database, Fabric, SQL, OAuth 2, Dimensional modeling, Data lakes, Data warehouses

## Experience

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### Senior Data Engineer Omniceil

September 2019 - Current  
Fort Worth, TX (remote)

- Established and enforced data reliability and observability standards by implementing monitoring, alerting, data quality validation frameworks, and SLA/SLO tracking to ensure high availability and trust in production data systems.
- Architected scalable, secure cloud-native data platforms with role-based access control, encryption, and governance best practices, supporting compliant and high-performance data access across enterprise environments.
- Built and orchestrated ETL pipelines delivering hundreds of GB of data daily from AWS into SQL Server-based data marts, improving data availability, freshness, and reporting speed.
- Managed large-scale cloud storage and backup solutions using AWS S3, Glacier, EBS, and lifecycle policies, overseeing 10+ TB of data and reducing storage costs by ~6–8% through intelligent tiering and retention strategies.
- Automated deployment and execution of benchmark, batch, and data-processing workloads using Docker, Ansible, and Git-based CI/CD pipelines, improving environment setup consistency and reducing configuration drift.
- Configured and managed Fivetran connectors to ingest data into Snowflake and support ELT pipelines and dbt transformations.
- Optimized Amazon Redshift for enterprise analytics workloads, enhancing ingestion throughput, schema design, and query performance for large BI datasets.
- Integrated AWS S3 into multiple production systems, enabling reliable, high-throughput access to raw and curated data for downstream analytics and reporting.
- Designed and scheduled Apache Airflow DAGs to orchestrate Hive, Pig, and Spark workflows with dependency management, retries, and SLA monitoring.
- Optimized Apache Spark / PySpark pipelines, improving job runtimes by ~6–9% through partitioning, caching, broadcast joins, and executor tuning.
- Collaborated with cross-functional teams to implement Talend-based ETL solutions, improving data consistency, error handling, and pipeline stability.
- Designed and implemented PL/SQL-based ETL processes to migrate, reconcile, and validate large datasets between heterogeneous systems with high data accuracy.
- Drove team delivery and code quality initiatives by participating in design reviews, establishing coding standards, and promoting best engineering practices.
- Contributed to the design of service-oriented and microservices-based components, improving deployment consistency and system maintainability.
- Applied established design patterns across multiple projects, improving code reuse, maintainability, and reducing regression issues.
- Developed Python-based REST APIs to capture, aggregate, and analyze system and application logs, improving monitoring visibility and reducing issue detection time.
- Implemented incremental models and materializations using dbt, enhancing transformation efficiency and resource utilization.

- Developed backend services using Python and Django to support high-volume transactional systems.
- Performed large-scale data analysis using Python, SQL, Hive, PySpark, and Spark SQL, improving processing throughput by ~6–8% through optimized joins and transformations.
- Delivered data solutions across AWS and Azure environments, improving scalability and reducing operational overhead.
- Built an automated data extraction framework using Python and Flask to integrate with Snowflake, Redis, Elasticsearch, MongoDB, and PostgreSQL.
- Containerized and deployed services using Docker and Kubernetes, improving deployment consistency and release stability.

**Technologies:** *Apache Spark / PySpark, Apache Airflow, AWS Data Platform (S3, Redshift, Glacier, EBS), Python, Talend, DBT, SQL / PL-SQL, Docker & Kubernetes, Data Marts & Schema Design, REST APIs (Flask, Django)*

## **Data Engineer**

January 2016 - August 2019

### **Gainwell Technologies**

Salt Lake City, UT

- Built and optimized batch data processing pipelines using PySpark on Databricks, improving job execution time by ~8–9% through partitioning, caching, and optimized joins.
- Implemented near-real-time ingestion pipelines using Apache Kafka, improving data delivery latency and downstream availability.
- Designed and implemented data models and schemas for Azure Data Lake Storage, improving query performance and reducing retrieval time.
- Automated ETL workflows using Talend, enabling reliable data movement between MongoDB and SQL Server.
- Used Apache NiFi for ingestion and flow-based processing, and Apache Airflow for scheduling and orchestration.
- Developed and maintained analytical data stores using Google BigQuery and Amazon Redshift, improving dashboard query performance by ~8–9%.
- Implemented backup, replication, and basic failover strategies for cloud data systems.
- Implemented log indexing and search using Elasticsearch, improving diagnostics and debugging workflows.
- Built RESTful backend services using Node.js and PHP, supporting high-traffic APIs.
- Developed and maintained Python-based APIs using Flask, SQLAlchemy, and PostgreSQL.

**Technologies:** *PySpark on Databricks, Apache Kafka, Apache Airflow, Talend, Apache NiFi, Google BigQuery, Amazon Redshift, Python (Flask, SQLAlchemy), Elasticsearch, Azure Data Lake Storage*

## **Software Engineer (Data & Web)**

July 2012 - December 2015

### **Surescripts LLC**

Arlington, VA

- Automated and managed AWS-based deployments using early Infrastructure-as-Code (CloudFormation) practices, improving environment consistency.
- Scaled Python-based web applications on AWS EC2 and Elastic Beanstalk, supporting growing user traffic.
- Optimized cloud storage and retrieval workflows on AWS.
- Supported application health using custom monitoring scripts, log analysis, and alerting.
- Led Agile-based development cycles, improving delivery predictability.
- Developed web applications using Python, Django, AngularJS (1.x), and Bootstrap.
- Managed full development lifecycle of multiple client projects.
- Built Python programs to retrieve and process data from AWS-hosted systems.
- Automated machine data ingestion and reporting using Python and SQL.
- Created analytical reports and visualizations using Python and Pandas.

**Technologies:** AWS EC2 & Elastic Beanstalk, Python, CloudFormation  
(Infrastructure-as-Code), Django, SQL, Pandas, Data Ingestion Automation, Monitoring & Log  
Analysis, AngularJS & Bootstrap, Agile Development

## Certifications

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- AWS Certified Data Analytics – Specialty
- Microsoft Certified: Azure Data Engineer Associate
- SnowPro Core & Advanced (Snowflake)
- Databricks Data Engineer

## Awards

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- Arctic Code Vault Contributor (Github)